

Project Backend Flow

1. Entry Point (server/src/server.ts)
 - Loads environment variables and database settings.
 - Connects to MongoDB using connectDatabase(DB_URI).
 - Starts the Express app on the configured PORT.
 - Verifies mail server connection after startup.
2. App Setup (server/src/app.ts)
 - Creates Express app instance.
 - Applies JSON body parser and cookie parser middleware.
 - Defines health check route at GET /.
 - Mounts main API routes under /api/v1/auth.
 - Uses a 404 handler for unknown routes.
 - Uses the global errorHandler middleware.
3. Main Routing (server/src/routes/index.ts)
 - Registers auth, user, brand, and category routers.
 - Each router defines its own endpoints and controllers.
 - Note: product.routes.ts is defined separately and should be imported into this file to make /api/v1/products available.
4. Product Routing (server/src/routes/product.routes.ts)
 - Defines product endpoints: GET /, GET /new_arrivals, GET /is_featured, GET /:id, GET /brand/:brandId, GET /category/:categoryId, POST /, PUT /:id, DELETE /:id.
 - Uses multer upload.fields() for cover_image and images.
 - Protects create, update, and delete with authenticate(All_Admins).
5. Product Controller (server/src/controllers/product.controller.ts)
 - create(): validates body fields and required cover_image.
 - Looks up brand and category by name before saving.
 - Uploads cover_image and images to Cloudinary via upload().
 - Saves the Product document with cover_image and images arrays.
 - getAllProducts(): returns all product documents.
 - getProductById(): returns a specific product by ID.
 - update(): updates the product document by ID.
 - deleteProduct(): removes media from Cloudinary and deletes the product.
6. Middleware and Utilities
 - auth.middlewares.ts handles user authentication and role checks.
 - multer.middlewares.ts handles file uploads from multipart/form-data.
 - errorHandler.middleware.ts catches and formats errors consistently.
 - cloudinary.utils.ts handles image upload and deletion to Cloudinary.
7. Data Flow for Create Product
 - Client sends POST /api/v1/products with form-data and files.
 - multer parses cover image+images into req files